

EEL5840 Fundamentals of Machine Learning

Spring 2024

Final Exam

May 2, 2024

Time Limit: 2 hours

Name: _____

- Write legibly.
- There are a total of 9 questions for a total of 100 points.
 - Some questions are worth more than other questions.
- **Closed-book, no computer, one-page formulas, calculator.**
 - **Write your name in the formula sheet.**

Answer the questions in the spaces provided on the question sheets. If you run out of room for an answer, continue on the back of the page.

Helpful Formulas

- True Positive Rate (TPR) and False Positive Rate (FPR) are defined as:

$$\text{TPR} = \frac{\text{TP}}{\text{TP} + \text{FN}}, \quad \text{FPR} = \frac{\text{FP}}{\text{FP} + \text{TN}}$$

Closed-book, no computer, one-page formulas, calculator

Grade Table (for the teaching team use only)

Question:	1	2	3	4	5	6	7	8	9	Total
Points:	10	10	10	10	10	15	15	10	10	100
Score:										

1. (10 points) Consider a binary classification task where each sample $\mathbf{x}_i$ is d -dimensional with its corresponding label $t_i \in \{-1, 1\}$. Suppose you have a dataset of N i.i.d. samples, $\{(x_i, t_i)\}_{i=1}^N$. Let $\mathbf{m}_k$ be the sample average for class k , and $\mathbf{m}$ the overall sample average. Suppose that your goal is to train a discriminant function of the form $y(\mathbf{x}) = \mathbf{w}^T \mathbf{x} + w_0$ that separates the two classes.

Suppose that you are interested in finding $\mathbf{w}$ and w_0 such that the **projected between-class variance is maximized**. Write down the objective function, the Lagrangian function, and derive for the analytical solutions for $\mathbf{w}$ (not w_0).

2. (10 points) Consider a support vector machine (SVM) classifier, $y(x) = \mathbf{w}^T \phi(x) + w_0$, for a two-class problem, $t \in \{-1, 1\}$, with the objective function,

$$\arg_{\mathbf{w}} \min \left\{ \frac{1}{2} \|\mathbf{w}\|^2 \right\} \quad \text{subject to} \quad t_n(\mathbf{w}^T \phi(\mathbf{x}_n) + w_0) \geq 1 \quad \forall n$$

Explain how does this objective function maximizes the margin of the discriminant function and avoids misclassified points.

3. (10 points) Consider the following 2-dimensional dataset, (x_1, x_2) , and its respective ground truth label, t :

x_1	x_2	t
1	0	1
4	2	1
0	-1	-1
-1	-1	-1
-2	1	-1

Consider the Perceptron classifier with the initial weights $w^{(0)} = [4, 1]$ and bias $w_0^{(0)} = 2$. Answer the following questions:

- (a) (3 points) Using the stochastic gradient descent, with a learning rate of $\eta = 0.1$, compute the new weights $w^{(1)}$ and the new bias $w_0^{(1)}$.

(b) (3 points) Compute another update for the weights $w^{(2)}$ and the bias $w_0^{(2)}$.

(c) (4 points) Repeat part (b) but using a momentum term with $\alpha = 0.9$.

4. (10 points) Suppose you have a training dataset with 10 samples $\{x_1, x_2, \dots, x_{10}\}$ and the associated target labels:

$$t = \{1, 2, 2, 3, 1, 2, 3, 3, 2, 1\}$$

Suppose you trained an MLP to produce a confidence of target given a sample. For the above data points, the binary classifier produced the following confidence values:

$$c = \{0.7, 0.6, 0.2, 0.3, 0, 0.9, 0.7, 0.4, 0.5, 0.99\}$$

Answer the following questions:

- (a) (6 points) Draw the associated ROC curve for class 2. Show your work.

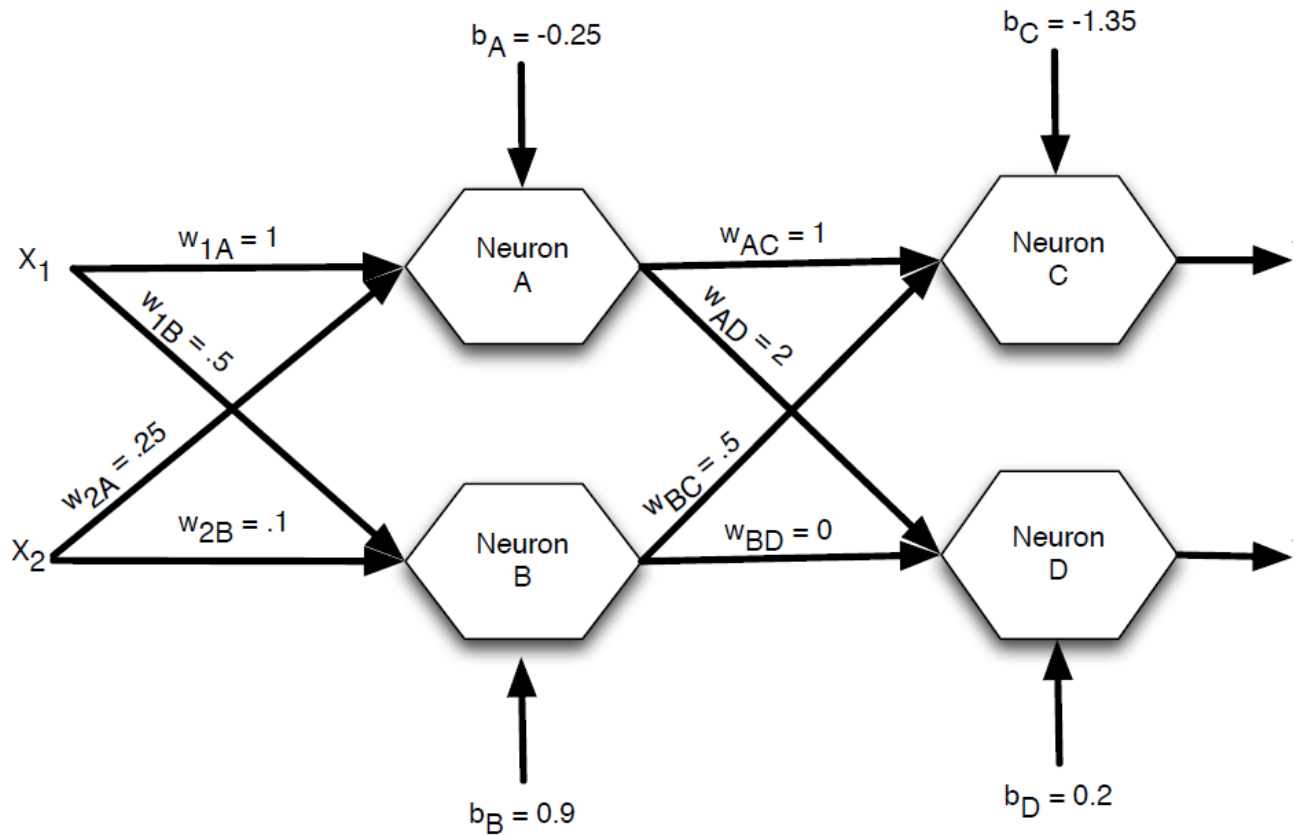
- (b) Suppose that the point of operation is a probability of 0.5, that is, a probability ≥ 0.5 is assigned to positive class (class 2), and probability < 0.5 is assigned to negative class (not class 2). What is the resulting confusion matrix? Show your work.

5. (10 points) Consider a feature matrix $\mathbf{X}$ of shape $D \times N$, where D is the dimensionality and N is the sample size (assume $N > D$). Assume that the mean of each feature is 0, that is, $\mu_{\mathbf{X}} = \mathbf{0}$. How is the covariance matrix of $\mathbf{X}$ utilized in PCA? What is the significance of eigenvalues and eigenvectors in PCA? How are they related to the principal components?

Solve for the principal components analytically to answer these questions.

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6. (15 points) This problem addresses Principal Component Analysis (PCA). Answer the following questions:
- (a) (5 points) How do you interpret the variance explained by each principal component?
 - (b) (5 points) What are the limitations of PCA? Discuss scenarios where PCA might not be suitable for dimensionality reduction.
 - (c) (5 points) How does PCA perform in the presence of multicollinearity among features? Does it mitigate or exacerbate the issue?

7. (15 points) Consider the following multi-layer perceptron (MLP) with 2 input units, 1 hidden layer with 2 units and 1 output layer with 2 units: All activation functions are



the ReLU activation function, $\phi(x) = \begin{cases} x, & x > 0 \\ 0, & x \leq 0 \end{cases}$ and $\phi'(x) = \begin{cases} 1, & x > 0 \\ 0, & x < 0 \end{cases}$.

Consider the following objective function,

$$J(\mathbf{w}) = \frac{1}{2} \sum_{i=1}^N (t_i - y_i)^2$$

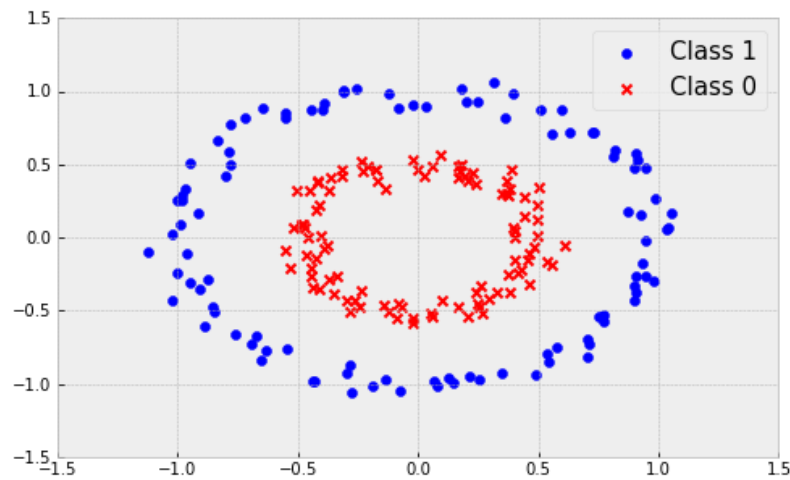
Consider the input sample $\mathbf{x} = [2, 1]^T$, with target $\mathbf{t} = [1.5, 4]^T$. Answer the following questions:

- (a) (3 points) Carry one forward pass for this training sample.

- (b) (6 points) Use backpropagation with standard gradient descent and a learning rate of $\eta = 0.01$ to update the weights w_{AC} and w_{AD} . Show your work.

- (c) (6 points) Use backpropagation (and your answers from part (b)) to update the weight w_{1A} . Show your work.

8. (10 points) Consider the following two-dimensional, two-class data set: Design an MLP



to obtain 100% classification accuracy on *this* data set. Be sure to define the network architecture and all parameters precisely (give exact numbers). Use the threshold activation function, $\phi(x) = \begin{cases} 1, & x > 0 \\ 0, & x \leq 0 \end{cases}$, for all units. Show and explain your work.

9. (10 points) This problem addresses the Multi-Layer Perceptron (MLP). Answer the following questions:
- (a) (4 points) How would you encode the output layer, its activation function, and which objective function would you choose to perform the following tasks: binary classification, multi-class classification, single-output regression and multiple-output regression?
 - (b) (2 points) Consider the feature map that results from convolving a $D \times D \times 3$ image a 3×3 filter. How many weight connections are there in this convolutional layer?
 - (c) (4 points) Identify at least two strategies to mitigate the occurrence of exploding gradients, and at least two strategies to avoid overfitting when training ANNs. Justify your choices.

HONOR STATEMENT

I understand that I am bound to uphold the honor code of the University of Florida. I have neither given nor received assistance on this examination. In addition, I did not use any outside materials on this exam other than the one page of formulas that was allowed.

Sign Your Name: _____

Write the Date: _____

Print Your Name: _____

Turn in your formula sheet with your exam!!!